

NexaTel — Data Dictionary

This document defines every table and every column in the NexaTel Customer Churn Analytics database. Use it as your reference while cleaning, joining, and analysing the data. The database contains 24 tables linked by primary and foreign keys.

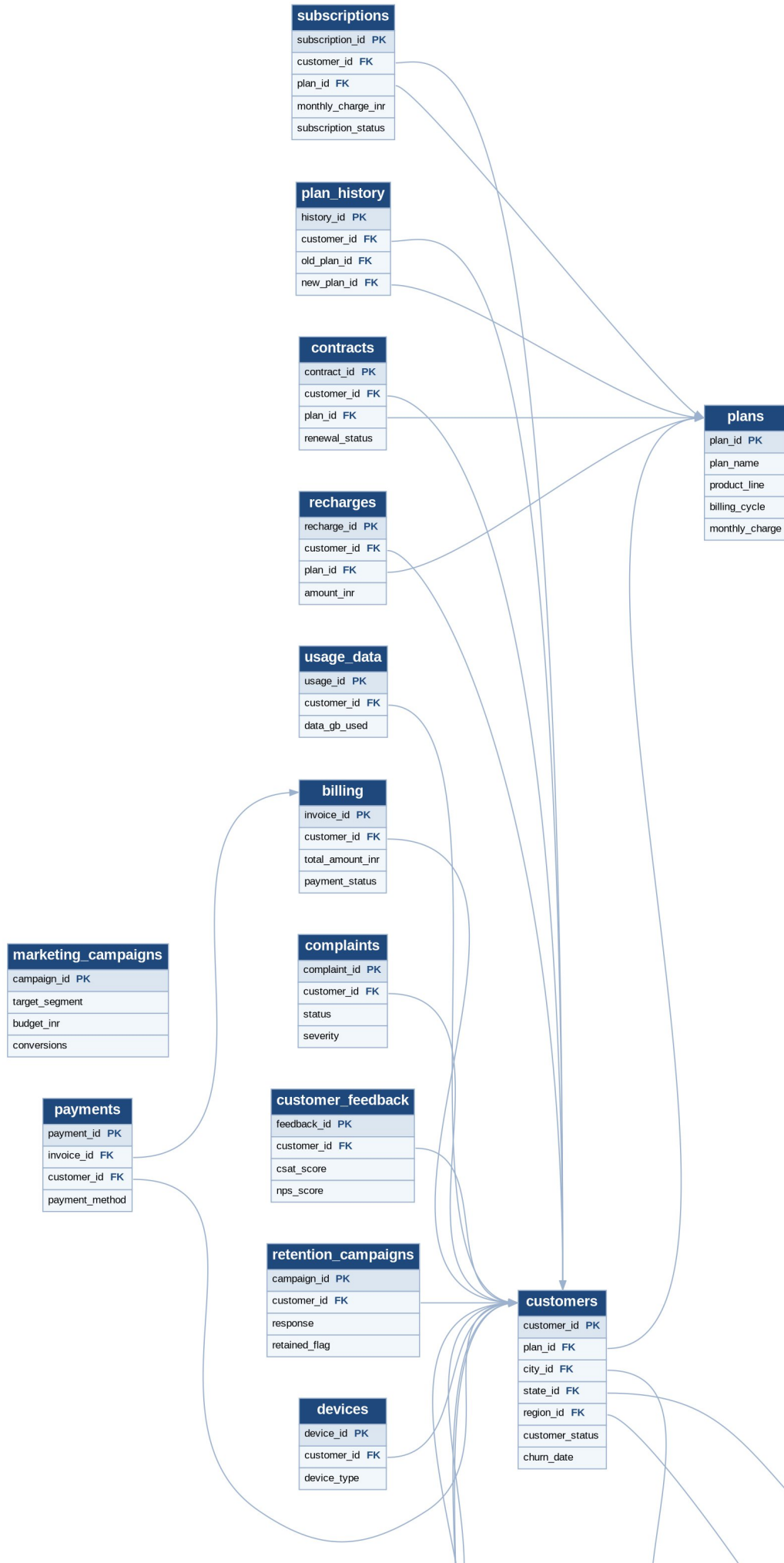
How to read this dictionary

For each table you will find a short purpose statement followed by a column table with these fields: **Column** (the field name), **Type** (storage type), **Key** (PK = primary key, FK = foreign key, — = none), **Null** (whether blanks are expected), **Example**, and **Description**. Date fields are stored as text in DD-MM-YYYY format.

Entity Relationship Diagram

The diagram below shows how the tables connect. The customers table is the hub — most tables join back to it through customer_id. A full-size version is provided as NexaTel_ERD.pdf.

NexaTel — Customer Churn Analytics : Entity Relationship Diagram



1. customers

One row per customer — the central table.

Column	Type	Key	Null	Example	Description
customer_id	VARCHAR	PK	No	C0000001	Unique customer identifier.
first_name	VARCHAR	—	No	Harish	Customer first name.
last_name	VARCHAR	—	No	Shetty	Customer last name.
gender	VARCHAR	—	No	Male	Male / Female.
date_of_birth	VARCHAR	—	No	31-03-1988	Date of birth (DD-MM-YYYY).
age	INT	—	No	38	Age in years (a few invalid values exist).
email	VARCHAR	—	Yes	harish.shetty909@yahoo.in	Email (some blank/invalid).
phone_number	VARCHAR	—	No	+916647481156	Mobile number (some invalid).
address	VARCHAR	—	Yes	Plot 382, Tilak Lane, Raipur	Postal address (some blank).
city_id	VARCHAR	FK	No	CT0026	Links to cities.city_id.
city_name	VARCHAR	—	No	Raipur	City name (some misspelt/upper-case).
state_id	VARCHAR	FK	No	ST027	Links to states.state_id.
pin_code	VARCHAR	—	No	492680	6-digit PIN (some invalid).
city_tier	VARCHAR	—	No	Tier-2	Metro / Tier-2 / Tier-3.
region_id	VARCHAR	FK	No	RG04	Links to regions.region_id.
occupation	VARCHAR	—	No	School Teacher	Customer occupation.
annual_income_inr	DECIMAL	—	Yes	791058.00	Estimated annual income (some blank).
customer_segment	VARCHAR	—	No	Mass	Mass / Value / Premium / Enterprise.
product_line	VARCHAR	—	No	Prepaid Mobile	Primary product line.
plan_id	VARCHAR	FK	Yes	PL002	Links to plans.plan_id.
acquisition_channel	VARCHAR	—	No	Retail Store	How the customer was acquired.
acquisition_date	VARCHAR	—	No	15-01-2026	Join date (DD-MM-YYYY; some other formats).
tenure_months	INT	—	No	2	Length of relationship in months.
customer_status	VARCHAR	—	No	Churned	Active / Churned — the analysis target.
churn_date	VARCHAR	—	Yes	22-03-2026	Date churned (blank if Active).

2. subscriptions

Subscriptions held by customers (active and terminated).

Column	Type	Key	Null	Example	Description
subscription_id	VARCHAR	PK	No	S0000001	Unique subscription ID.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
plan_id	VARCHAR	FK	Yes	PL002	Links to plans.plan_id (a few null).

start_date	VARCHAR	—	No	15-01-2026	Subscription start (DD-MM-YYYY).
end_date	VARCHAR	—	Yes	22-03-2026	End date (blank if active).
billing_cycle	VARCHAR	—	No	Monthly	Monthly / Annual.
monthly_charge_inr	DECIMAL	—	No	299.00	Recurring charge in INR.
subscription_status	VARCHAR	—	No	Terminated	Active / Terminated.

3. plans

Catalogue of available plans (lookup table).

Column	Type	Key	Null	Example	Description
plan_id	VARCHAR	PK	No	PL001	Unique plan ID.
plan_name	VARCHAR	—	No	Prepaid Saver 199	Marketing plan name.
product_line	VARCHAR	—	No	Prepaid Mobile	Product line of the plan.
billing_cycle	VARCHAR	—	No	Monthly	Monthly / Annual.
monthly_charge	DECIMAL	—	No	199	Monthly charge in INR.
data_quota_gb	INT	—	No	28	Monthly data quota (-1 = unlimited).
voice_minutes	INT	—	No	-1	Voice allowance (-1 = unlimited).
sms_count	INT	—	No	300	SMS allowance (-1 = unlimited).
segment	VARCHAR	—	No	Mass	Target segment for the plan.
is_active	VARCHAR	—	No	Yes	Whether the plan is currently offered.

4. plan_history

Record of plan upgrades / downgrades over time.

Column	Type	Key	Null	Example	Description
history_id	VARCHAR	PK	No	PH0000001	Unique change-event ID.
customer_id	VARCHAR	FK	No	C0000005	Links to customers.customer_id.
old_plan_id	VARCHAR	FK	No	PL003	Plan before the change.
new_plan_id	VARCHAR	FK	No	PL010	Plan after the change.
change_date	VARCHAR	—	No	29-10-2019	Date of change (DD-MM-YYYY).
change_reason	VARCHAR	—	No	Customer Request	Reason for the plan change.

5. contracts

Contracts for postpaid / fibre / enterprise customers.

Column	Type	Key	Null	Example	Description
contract_id	VARCHAR	PK	No	CN000001	Unique contract ID.
customer_id	VARCHAR	FK	No	C0000002	Links to customers.customer_id.
plan_id	VARCHAR	FK	No	PL017	Links to plans.plan_id.
contract_type	VARCHAR	—	No	Fiber Broadband	Type of contract.
contract_length_months	INT	—	No	24	Contract length in months.
start_date	VARCHAR	—	No	01-01-2014	Contract start (DD-MM-YYYY).
end_date	VARCHAR	—	No	22-12-2015	Contract end (DD-MM-YYYY).

auto_renew	VARCHAR	—	No	Yes	Whether auto-renewal is on.
renewal_status	VARCHAR	—	No	Renewed	Renewed / Auto-Renewed / Pending / Not Renewed / Cancelled.

6. devices

Device held by each customer.

Column	Type	Key	Null	Example	Description
device_id	VARCHAR	PK	No	DEV0000001	Unique device ID.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
brand	VARCHAR	—	No	Voltic	Device brand.
model	VARCHAR	—	No	V30 5G	Device model.
device_type	VARCHAR	—	No	Smartphone	Smartphone / Router / IoT Device.
device_price_inr	DECIMAL	—	No	21999	Device price in INR.
purchase_date	VARCHAR	—	No	02-02-2026	Purchase date (DD-MM-YYYY).
imei	VARCHAR	—	No	505331931165643	Device IMEI number.

7. billing

Monthly invoices for billed (non-prepaid) customers.

Column	Type	Key	Null	Example	Description
invoice_id	VARCHAR	PK	No	INV00000001	Unique invoice ID.
customer_id	VARCHAR	FK	No	C0000002	Links to customers.customer_id.
billing_date	VARCHAR	—	No	31-03-2026	Invoice date (a few future dates).
billing_period_month	VARCHAR	—	No	03-2026	Billing month (MM-YYYY).
base_amount_inr	DECIMAL	—	No	1490.77	Charge before GST.
gst_amount_inr	DECIMAL	—	No	268.34	18% GST (some incorrect).
total_amount_inr	DECIMAL	—	No	1759.11	Total payable (a few negative).
due_date	VARCHAR	—	No	15-04-2026	Payment due date.
payment_status	VARCHAR	—	No	Paid	Paid / Unpaid / Overdue.

8. payments

Payments made against invoices.

Column	Type	Key	Null	Example	Description
payment_id	VARCHAR	PK	No	PAY00000001	Unique payment ID.
invoice_id	VARCHAR	FK	Yes	INV00000001	Links to billing.invoice_id (a few broken).
customer_id	VARCHAR	FK	No	C0000002	Links to customers.customer_id.
payment_date	VARCHAR	—	No	04-04-2026	Date of payment.
amount_inr	DECIMAL	—	No	1759.11	Amount paid in INR.
payment_method	VARCHAR	—	No	Debit Card	UPI / Credit Card / ... / RTGS.
payment_status	VARCHAR	—	No	Success	Payment outcome.

9. recharges

Prepaid recharge transactions.

Column	Type	Key	Null	Example	Description
recharge_id	VARCHAR	PK	No	RC00000001	Unique recharge ID.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
recharge_date	VARCHAR	—	No	18-01-2026	Recharge date (DD-MM-YYYY).
amount_inr	DECIMAL	—	No	303.90	Recharge amount in INR.
plan_id	VARCHAR	FK	No	PL002	Links to plans.plan_id.
payment_method	VARCHAR	—	No	UPI	Payment method used.

10. usage_voice

Monthly voice usage per customer.

Column	Type	Key	Null	Example	Description
usage_id	VARCHAR	PK	No	UV00000001	Unique usage record.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
usage_month	VARCHAR	—	No	03-2026	Usage month (MM-YYYY).
voice_minutes_used	INT	—	No	439	Minutes used in the month.
outgoing_calls	INT	—	No	356	Outgoing call count.
incoming_calls	INT	—	No	255	Incoming call count.

11. usage_sms

Monthly SMS usage per customer.

Column	Type	Key	Null	Example	Description
usage_id	VARCHAR	PK	No	US00000001	Unique usage record.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
usage_month	VARCHAR	—	No	03-2026	Usage month (MM-YYYY).
sms_sent	INT	—	No	19	SMS sent in the month.
sms_received	INT	—	No	20	SMS received in the month.

12. usage_data

Monthly data usage per customer.

Column	Type	Key	Null	Example	Description
usage_id	VARCHAR	PK	No	UD00000001	Unique usage record.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
usage_month	VARCHAR	—	No	03-2026	Usage month (MM-YYYY).
data_gb_used	DECIMAL	—	No	22.35	Data used in GB (a few outliers).
data_sessions	INT	—	No	1047	Number of data sessions.

13. network_quality

Monthly network experience per customer.

Column	Type	Key	Null	Example	Description
record_id	VARCHAR	PK	No	NQ00000001	Unique record ID.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
city_id	VARCHAR	FK	No	CT0026	Links to cities.city_id.
month	VARCHAR	—	No	03-2026	Month (MM-YYYY).
avg_signal_strength_dbm	DECIMAL	—	No	-74.3	Average signal strength (dBm).
dropped_call_rate_pct	DECIMAL	—	No	3.15	Dropped-call rate (%).
avg_download_speed_mbps	DECIMAL	—	No	25.1	Average download speed (Mbps).
network_downtime_hours	DECIMAL	—	No	0.87	Network downtime hours in the month.

14. support_tickets

Support tickets raised by customers.

Column	Type	Key	Null	Example	Description
ticket_id	VARCHAR	PK	No	TK00000001	Unique ticket ID.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
created_date	VARCHAR	—	No	08-09-2017	Ticket creation date.
category	VARCHAR	—	No	Billing Dispute	Issue category.
priority	VARCHAR	—	No	Medium	Low / Medium / High / Critical.
channel	VARCHAR	—	No	Store	Contact channel.
status	VARCHAR	—	No	Pending	Resolved / Pending / Escalated / Reopened.
first_contact_resolution	VARCHAR	—	No	No	Yes / No — solved on first contact.
assigned_employee_id	VARCHAR	FK	No	EM00222	Links to employees.employee_id.
resolution_hours	DECIMAL	—	Yes	36.5	Hours to resolve (blank if open).
resolution_date	VARCHAR	—	Yes		Date resolved (blank if open).

15. complaints

Formal complaints raised by customers.

Column	Type	Key	Null	Example	Description
complaint_id	VARCHAR	PK	No	CMP00000001	Unique complaint ID.
customer_id	VARCHAR	FK	No	C0000001	Links to customers.customer_id.
complaint_date	VARCHAR	—	No	30-06-2020	Complaint date.
complaint_type	VARCHAR	—	No	Data Speed	Type of complaint.
severity	VARCHAR	—	No	Major	Minor / Moderate / Major / Severe.
status	VARCHAR	—	No	Unresolved	Resolved / Unresolved — a churn driver.
resolution_date	VARCHAR	—	Yes		Date resolved (blank if unresolved).

16. customer_feedback

CSAT / NPS survey responses.

Column	Type	Key	Null	Example	Description
feedback_id	VARCHAR	PK	No	FB00000001	Unique feedback ID.
customer_id	VARCHAR	FK	No	C0000002	Links to customers.customer_id.
feedback_date	VARCHAR	—	No	13-03-2025	Survey date.
csat_score	INT	—	No	5	Satisfaction score (1-5).
nps_score	INT	—	No	7	Net Promoter score (0-10).
channel	VARCHAR	—	No	App Rating	Survey channel.
feedback_category	VARCHAR	—	No	Network	Topic of the feedback.

17. retention_campaigns

Customer-level retention offers and outcomes.

Column	Type	Key	Null	Example	Description
campaign_id	VARCHAR	PK	No	RTC0000001	Unique retention-contact ID.
customer_id	VARCHAR	FK	No	C0000004	Links to customers.customer_id.
campaign_name	VARCHAR	—	No	Renewal Nudge	Retention campaign name.
offer_type	VARCHAR	—	No	Annual Plan Discount	Offer presented.
contact_date	VARCHAR	—	No	04-09-2025	Date of contact.
channel	VARCHAR	—	No	SMS	Contact channel.
response	VARCHAR	—	No	Declined	Accepted / Declined / No Response.
retained_flag	VARCHAR	—	No	Yes	Whether the customer was retained.

18. marketing_campaigns

Campaign-level marketing definitions.

Column	Type	Key	Null	Example	Description
campaign_id	VARCHAR	PK	No	MKT00001	Unique campaign ID.
campaign_name	VARCHAR	—	No	Festive Recharge Fest 2025	Campaign name.
channel	VARCHAR	—	No	In-App	Primary channel.
target_segment	VARCHAR	—	No	High-Risk Churn	Targeted segment.
start_date	VARCHAR	—	No	16-04-2025	Campaign start.
end_date	VARCHAR	—	No	28-06-2025	Campaign end.
budget_inr	DECIMAL	—	No	20354972	Campaign budget in INR.
customers_reached	INT	—	No	250093	Customers reached.
conversions	INT	—	No	8729	Conversions achieved.

19. employees

Support / sales staff who handle tickets.

Column	Type	Key	Null	Example	Description
employee_id	VARCHAR	PK	No	EM00001	Unique employee ID.
employee_name	VARCHAR	—	No	Asha Sharma	Employee name.

gender	VARCHAR	—	No	Female	Gender.
department	VARCHAR	—	No	Billing	Department.
role	VARCHAR	—	No	Support Agent	Job role.
store_id	VARCHAR	FK	No	SR0176	Links to stores.store_id.
region_id	VARCHAR	FK	No	RG04	Links to regions.region_id.
hire_date	VARCHAR	—	No	31-01-2021	Hire date.
email	VARCHAR	—	No	asha.sharma@nexatel.co.in	Work email.

20. stores

Retail stores by city and region.

Column	Type	Key	Null	Example	Description
store_id	VARCHAR	PK	No	SR0001	Unique store ID.
store_name	VARCHAR	—	No	NexaTel Gaya Point	Store name.
city_id	VARCHAR	FK	No	CT0057	Links to cities.city_id.
state_id	VARCHAR	FK	No	ST024	Links to states.state_id.
region_id	VARCHAR	FK	No	RG04	Links to regions.region_id.
store_type	VARCHAR	—	No	Experience Centre	Store type.
opened_date	VARCHAR	—	No	19-10-2017	Date the store opened.

21. regions

Operating zones (lookup table).

Column	Type	Key	Null	Example	Description
region_id	VARCHAR	PK	No	RG01	Unique region ID.
region_name	VARCHAR	—	No	North	North / West / South / East.
zone_hq_city	VARCHAR	—	No	Gurugram	Zone headquarters city.

22. cities

Cities with tier and PIN prefix (lookup table).

Column	Type	Key	Null	Example	Description
city_id	VARCHAR	PK	No	CT0001	Unique city ID.
city_name	VARCHAR	—	No	New Delhi	City name.
state_id	VARCHAR	FK	No	ST001	Links to states.state_id.
state_name	VARCHAR	—	No	Delhi	State name.
zone	VARCHAR	—	No	North	Operating zone.
region_id	VARCHAR	FK	No	RG01	Links to regions.region_id.
tier	VARCHAR	—	No	Metro	Metro / Tier-2 / Tier-3.
pin_prefix	VARCHAR	—	No	110	First 3 digits of the PIN.

23. states

Indian states and union territories (lookup table).

Column	Type	Key	Null	Example	Description
state_id	VARCHAR	PK	No	ST001	Unique state ID.
state_name	VARCHAR	—	No	Delhi	State / UT name.
state_code	VARCHAR	—	No	DL	2-letter code.
zone	VARCHAR	—	No	North	Operating zone.

24. data_quality_issue_log

Reference list of data-quality issues to fix during cleaning.

Column	Type	Key	Null	Example	Description
issue_table	VARCHAR	—	No	customers	Table the issue affects.
issue_type	VARCHAR	—	No	missing_email	Type of issue.
rows_affected	INT	—	No	760	Number of rows affected.